Low-Order Decision Certificates and Value-Estimation Limits in Structured Quantum Compilation

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Abstract

Low-order statistics can decide whether optimization is needed while failing to determine an optimum's value or structure. We prove this hierarchy for a Pauli partition compiler with a fixed structural block-encoding objective. For every $m \geq 5$, the unary compiler is globally optimal exactly when every pair gain is nonpositive and the sum of any two disjoint pair gains plus one is nonpositive. The largest clause touches four term indices, and an explicit $m = 4$ instance proves the threshold sharp.

For every $t \geq 1$, two explicit $5t$-term families have identical ordered weights and complete labeled pair-gain matrices, yet their exact improvements are $12t - 2$ and $10t - 1$. Pair-only estimators therefore incur real additive error at least $(2t - 1)/2$, integer error at least t, and symmetric factor at least $\sqrt{(12t - 2)/(10t - 1)}$; no uniform factor below $\sqrt{6/5}$ is possible. One family forces a triple block, while the other uses only pairs and singletons.

For every $m \geq 5$ and $L \geq 1$, a second pair of instances agrees on every labeled common-factor count through order $m-2$ but has value gap $[m(\lceil \log_2 m \rceil + 1) - 1]L$. Möbius inversion proves that the parity trade is the unique nonzero integer direction invisible to all proper labeled marginals. These are representation limits, not computational-hardness claims.

Keywords: quantum compilation; low-order certificates; information lower bounds; minimax estimation; Markov bases; Möbius inversion

1. Introduction

Global combinatorial optimization supports several distinct downstream questions: whether a baseline is optimal, how much improvement is available, what structure an optimizer must contain, and whether a feature representation approximates the optimum uniformly. A representation can be complete for one question and incomplete for another.

We give an exact scalable example in structured quantum compilation. The compiler partitions m Pauli terms into arbitrary blocks, extracts common Pauli factors, and pays a structural cost. Although its candidate space contains every set partition, unary optimality is decided by inequalities involving at most four indices. This makes pair information appear unusually powerful. Our remaining theorems locate its exact limits.

The contributions are:

1. a four-index decision theorem for every $m \geq 5$, with an exact four-term counterexample;

2. two pair-indistinguishable product families with unbounded additive value gap and forced triple-versus-pair optimizer structure;
3. exact real, integer, symmetric multiplicative, and one-sided minimax lower bounds;
4. a construction showing that interactions through order $m-2$ do not determine exact value; and
5. a primitive-kernel theorem showing that the corresponding invisible difference trade must be the dense parity direction.

2. Compiler and objective

An instance is an ordered tuple of nonidentity Pauli strings $p_1, \ldots, p_m$. Let w_i be the weight of term i, let $W = \sum_i w_i$, write $w(S) = \sum_{i \in S} w_i$, and let $f(S)$ count the columns on which every term in a nonempty block S has the same nonidentity Pauli.

The compiler chooses a set partition Π. We define the fixed equal-weight structural objective directly. For $|\Pi| \geq 2$, its cost is

$$C(\Pi) = 2m + |\Pi| - 3 + \sum_{S \in \Pi} d(|S|) + \max_{S \in \Pi} b(|S|) + \sum_{S \in \Pi} \left[2f(S) + (b(|S|) + 2)(w(S) - |S| f(S))\right],$$

where $b(s) = \lceil \log_2 s \rceil$, $b(1) = 0$, and

$$d(1) = 0, \qquad d(s) = d(\lceil s/2 \rceil) + d(\lfloor s/2 \rfloor) + s - 2.$$

The unary incumbent has cost $C_U = 2W + 3m - 3$. For the one-block partition, the declared flag convention gives

$$C_{\text{one}} = (b(m) + 1)W + m - 1 + d(m) + b(m) - [m(b(m) + 1) - 1] f([m]).$$

Together these equations completely define the mathematical compiler studied below; no unstated dominance claim is used. These are structural compiler costs, not physical T-gate counts, depth, runtime, qubits, or fault-tolerant overhead.

3. A four-index decision certificate

For every pair, define

$$g_{ij} = 4f(\{i, j\}) - (w_i + w_j).$$

Let $\mathcal{P}_4(m)$ require

$$g_{ij} \leq 0$$

for every pair and

$$g_{ij} + g_{k\ell} + 1 \le 0$$

for every two disjoint pairs.

Theorem 1. For every $m \ge 5$,

$$\min_{\Pi} C(\Pi) = C_U \quad \Longleftrightarrow \quad \mathcal{P}_4(m).$$

Proof. First suppose $|\Pi| \ge 2$. For a block S of size s, set

$$T_s(S) = [s(b(s)+2) - 2]f(S) - b(s)w(S), \qquad h_s = s - 1 - d(s).$$

Direct expansion of the cost gives

$$C_U - C(\Pi) = \sum_{S \in \Pi} \left(T_{|S|}(S) + h_{|S|}\right) - \max_{S \in \Pi} b(|S|).$$

The depth recurrence gives $h_s = 3 - s + h_{\lceil s/2 \rceil} + h_{\lfloor s/2 \rfloor}$, and hence $h_2 = h_3 = h_4 = 1$, $h_5 = 0$, and $h_s \le 0$ for $s \ge 5$. For a block of size $s \ge 3$, take a perfect matching if s is even and an s-cycle if s is odd. Summing the pair clauses counts every term once or twice and uses $f(\{i,j\}) \ge f(S)$, giving $w(S) \ge 2sf(S)$. Therefore

$$T_s(S) \le [s(2 - b(s)) - 2]f(S) \le -2;$$

when $f(S) = 0$, the stronger bound $T_s(S) = -b(s)w(S) \le -b(s)s$ applies. Thus every block of size at least three has total credit $T_s + h_s \le -1$.

For a pair, $T_2 + h_2 = g_{ij} + 1$. Pair blocks in one partition are disjoint. Their gains are integers, every $g_{ij} \le 0$, and the two-pair clause prevents two of them from being zero. Hence any collection of pair blocks has total credit at most one. If the partition contains only pairs and singletons, the final width term is one and the total gain is nonpositive. If it contains a larger block, that block cancels the possible unit of pair credit and the width term is at least two, again making the gain nonpositive.

For the exceptional one-block partition, write $b = b(m)$, $F = f([m])$, and $d = d(m)$. Direct calculation gives

$$C_U - C_{\text{one}} = (1-b)W + [m(b+1) - 1]F + 2m - 2 - d - b.$$

The same matching/cycle sum gives $W \ge 2mF$. If $F = 0$, then $W \ge m$ and the gain is negative. If $F \ge 1$, the recurrence closes $5 \le m \le 8$ with $b = 3, d = 2m - 6$, closes $9 \le m \le 16$ with $b = 4, d \ge m - 1$, and $b \ge 5$ makes the remaining coefficient negative for $m \ge 17$. Thus no partition improves on unary when $\mathcal{P}_4(m)$ holds.

Conversely, a failed pair clause is witnessed by that pair and all remaining singletons; a failed disjoint-pair clause is witnessed by those two pairs and the remaining singletons. Their gains are respectively g_{ij} and $g_{ij} + g_{k\ell} + 1$. $\square$

The four-term instance

$$XXII, \quad XYII, \quad XZII, \quad XIXX$$

satisfies both clause families, yet has $C_U = 27$ and one-block cost 23. The term-count threshold is therefore sharp.

4. Complete pair information, different values and optimizers

The local five-term gadgets act on six qubits. Written with the first qubit on the left, they are

$$\begin{aligned} A_1 &= (\mathrm{XXXXII}, \mathrm{XXXIXI}, \mathrm{XXXIIX}, \mathrm{XXIIII}, \mathrm{XXIIII}), \\ B_1 &= (\mathrm{XXXXII}, \mathrm{XXXIXI}, \mathrm{XXIXXI}, \mathrm{XXIIII}, \mathrm{XXIIII}). \end{aligned}$$

Both have ordered weights $(4, 4, 4, 2, 2)$, and direct intersection of their supports gives the same common-factor count for every labeled pair. For $t \geq 1$, let A_t and B_t be t copies on disjoint six-qubit coordinate sets.

To determine their optima, use the block credit $U(S) = T_{|S|}(S) + h_{|S|}$ from the proof of Theorem 1. Enumerating the 31 nonempty subsets and 52 partitions of one five-term gadget gives a unique maximum $\sum U = 12$ for A_1, at the triple-plus-pair partition $\{\{1, 2, 3\}, \{4, 5\}\}$. For B_1, the maximum of $\sum U$ is 10. Besides pair-and-singleton partitions, three triple-plus-pair partitions also attain this credit maximum: $\{\{1, 4, 5\}, \{2, 3\}\}$, $\{\{1, 3\}, \{2, 4, 5\}\}$, and $\{\{1, 2\}, \{3, 4, 5\}\}$. Their width penalty is two, so their improvement is eight; a pair-and-singleton maximizer pays width one and improves by nine. Consequently every cost-optimal B_1 partition uses only pairs and singletons. A block meeting two gadgets has $f(S) = 0$ and strictly negative credit. Replacing every such mixed block by singleton blocks changes its credit to zero and cannot increase the maximum index width. Repeating this replacement removes all mixed blocks, so the global optimum therefore decomposes by gadgets. Accounting once for the shared maximum-width term yields

$$\Delta_A(t) = 12t - 2, \qquad \Delta_B(t) = 10t - 1.$$

Their value gap is $2t - 1$. Every optimum in A_t contains the distinguished triple block and one pair in each gadget. Every optimum in B_t uses only pairs and singletons. Thus complete pair information determines neither the exact improvement nor the presence of a triple block in an optimizer. The ancillary verifier reproduces every local subset/partition row and performs direct full-partition checks for $t = 1, 2$; the scalable formulas follow from the displayed decomposition.

5. Exact minimax consequences

Let Φ be any deterministic real-valued estimator whose input is exactly the term count, ordered weights, and complete labeled pair-gain matrix. The two fiber members have identical inputs, so Φ returns one value y_t for both.

Theorem 2 (additive minimax radius). For every $t \geq 1$,

$$\max\{|y_t-\Delta_A|,|y_t-\Delta_B|\}\geq \frac{2t-1}{2}.$$

The midpoint attains equality. If the estimator must return an integer, the minimum worst-case error is exactly t.

Proof. Two real values at distance $2t-1$ cannot both lie within a smaller radius of one estimate. The integer radius is the ceiling of half the distance. □

5.1 Symmetric multiplicative estimation

For positive improvements, use the convention

$$\Delta/\rho \leq y \leq \rho\Delta, \qquad \rho\geq 1.$$

Theorem 3. A common estimate valid for both fiber members requires

$$\rho \geq \sqrt{\frac{\Delta_A}{\Delta_B}}=\sqrt{\frac{12t-2}{10t-1}}.$$

The geometric mean attains equality. The bound increases with t and tends to $\sqrt{6/5}$. Hence no estimator using only the stated pair information has a uniform symmetric factor strictly below $\sqrt{6/5}$ on this family.

For one-sided certificates, a common upper estimate must be at least Δ_A but at most $\alpha\Delta_B$ to approximate B_t; the same ratio applies to a common lower estimate. Thus the asymptotic one-sided factor is at least $6/5$. These are information lower bounds, not computational hardness results.

6. Proper interactions still miss exact value

Fix $m\geq 5$, set $q=m-1$, $b=\lceil\log_2 m\rceil$, and distinguish one anchor from q variable terms. For every subset $S\subseteq[q]$ of one parity, place L columns supported on the anchor together with the variables in S; use the opposite parity in the second instance. Each side therefore has $N=2^{m-2}L$ trade columns. Add to both instances

$$K=Nm(b+1)+m-1+d(m)+b+1$$

columns supported on all m terms.

The resulting pair agrees on ordered weights and on every labeled common-factor count for subsets of size at most $m-2$, yet

$$\Delta(A)-\Delta(B)=[m(\lceil\log_2 m\rceil+1)-1]\,L.$$

Proof. For every labeled subset of terms of size at most $m-2$, the two parity classes contain the same number of supersets, so all common-factor counts through order $m-2$ agree. The common padding also makes the one-block partition uniquely optimal: for any proper partition, the exact cost difference from the full block is bounded below by

$$3K - Nm(b+1) - [m - 1 + d(m) + b] > 0.$$

Only the full common-factor count differs. Its coefficient in the one-block improvement is $1-m(b+1)$, while the parity imbalance is L, giving the displayed absolute gap. Thus, for fixed m, the ambiguity is unbounded in L. $\square$

7. The unique invisible direction

Represent a trade column by a subset of $[q]$. Let $\delta : 2^{[q]} \to \mathbb{Z}$ be the signed multiplicity difference, and define its upper marginal by

$$M(T) = \sum_{S \supseteq T} \delta(S).$$

Equality of all proper labeled marginals is $M(T) = 0$ for every proper $T \subsetneq [q]$.

Theorem 4 (proper-marginal kernel). If every proper upper marginal vanishes, then

$$\delta(S) = (-1)^{q-|S|}c, \qquad c = \delta([q]).$$

Proof. Möbius inversion on the Boolean lattice expresses $\delta(S)$ as the alternating sum of upper marginals over supersets of S. Only the top marginal can remain. $\square$

If $c \neq 0$, every Boolean cell is used and exactly half occur on each signed side. A primitive integer trade therefore has mass at least $2^{q-1} = 2^{m-2}$ on each side. The parity construction with $L = 1$ attains this bound. The statement proves minimality of the *difference trade*; it does not prove that the common padding is minimal.

8. Relation to prior work

Markov-basis and hierarchical-model theory [1-5] owns the generic language of fibers, marginal-preserving moves, toric ideals, and higher-order interactions invisible to lower marginals. Boolean-lattice Möbius inversion is classical.

The residual result is the exact conjunction for the fixed compiler objective: a four-index certificate decides global unary optimality for all $m \geq 5$, the same pair representation has exact unbounded and multiplicative value limits, optimizer block structure is not identifiable, and even order-$m-2$ information misses exact value through the primitive parity direction.

9. Reproducibility and limitations

The four-term witness, pair-indistinguishable gadgets, product formulas, minimax optima, and Boolean-lattice kernel have been checked with separate exact implementations; the analytic arguments provide all-parameter authority.

The results apply to the stated structural objective. The decision theorem does not reconstruct an optimizer. The minimax bounds apply to estimators restricted to the exact declared representation. The high-order construction uses conservative common padding whose minimality is open. The work proves information nonidentifiability, not a complexity-class lower bound. Transfer to other compiler grammars or physical resource models remains open.

10. Conclusion

Information sufficiency is query-dependent. Complete pair information decides whether the unary compiler is optimal, yet it cannot recover the improvement value within arbitrarily small additive or fixed multiplicative error and cannot determine optimizer block structure. Even all labeled interactions below the top two orders fail to determine exact value, and the invisible integer direction is necessarily dense.

A perfect decision statistic is therefore not automatically a useful value statistic. In this compiler, the distinction is exact, scalable, and independent of computational assumptions.

Data and code availability

Exact implementations reproducing the four-term witness, paired gadget families, product formulas, minimax calculations, and Boolean-lattice kernel accompany the submission source. They are verification aids; the displayed arguments carry the all-parameter claims. These files are distributed in the source archive accompanying this version.

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